

Memory Aid Design and Secondary Risk (U.S. Air Traffic Control)

Seed Log (Retrospective Seed – SL-003)

Timeframe; Recorded : Winter 2026; March 2026

Revision Note: Formatting and timeframe standardization updated (Version 1.1)

Observation: A memory aid introduced to the tower environment appeared to reduce the time and focus operators devoted to scanning the airport surface and surrounding airspace. The tool was intended to assist controllers in remembering when aircraft were crossing active runways. Some operators occasionally stopped using the memory aid during periods of higher traffic volume.

Trigger: While reading about human factors and safety management systems in aviation, the operator encountered the concept that tools designed to mitigate operational risk may also introduce secondary risks.

V.A.I.Y.L. Sequence

Variation

The operator noticed that both themselves and other controllers did not always maintain the expected or optimal visual scan out the tower windows when using the memory aid.

Analysis

According to facility management, the memory aid had been introduced as a corrective action following a quality review of an operational incident. The operator understood that the tool was intended to reduce the likelihood of errors associated with aircraft crossing active runways.

Note: The operator first noticed the potential effect of the memory aid approximately two years earlier when the tool was introduced. During the winter of 2026, after reading about risk mitigation tools and secondary risk in safety management and human factors literature, the operator discussed the concept with a trainee and reconsidered how it might apply to the facility's crossing runway memory aid.

Examination (external to the framework)

Notes: N/A

Operational Process: N/A

Operational Review: N/A

Insight

The operator inferred that the memory aid may have been designed primarily to satisfy a corrective action requirement following the earlier incident, with limited consideration given to how the tool might affect operator attention. Because the memory aid is positioned low on the tower control panel below the windows, its use may reduce the time controllers visually scan the operational environment. Additionally, the facility's SOPs require the memory aid to be used but do not specify a standardized method for its use.

Yield

The memory aid may introduce a form of secondary risk within tower operations by drawing operator attention away from direct observation of the operational environment.

Learn

The operator recognized the need to balance the use of procedural tools with maintaining situational awareness of the operational environment. Increased familiarity with safety management and human factors concepts, including secondary risk, provided a clearer framework for understanding how safety tools may influence operator behavior.

Scope

This specific memory aid is unique to the facility. However, because crossing runway configurations are common across air traffic control facilities, similar memory aids elsewhere could introduce comparable effects on operator attention and visual scanning.